

Bayesian Learning & MonteCarlo Simulation

Hourly Load Factor - Final Project

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Introduction

Task 1: Beta Mixture & MCMC Diagnostics

Task 2: Posterior Predictive Checks

Task 3: Post-Hoc Clustering

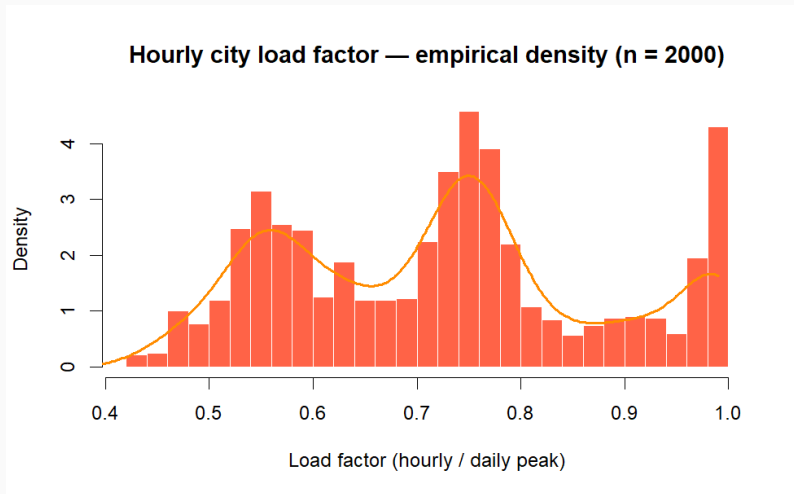
Task 4: Temporal Modeling

Introduction

- **Dataset:** Hourly power consumption in Tetouan, Morocco (2017).
- **Target Variable:** Hourly Load Factor $LF \in (10^{-3}, 1 - 10^{-3})$.
- **Objective:** Model the complex, multimodal distribution of the load factor using Beta distributions and analyze its temporal dynamics.
- **Table:**

Date	Hour	LF
------	------	----

Load Factor Distribution



Task 1: Beta Mixture & MCMC Diagnostics

Likelihood

$H = 1 : y_i \sim \text{Beta}(\alpha, \beta)$

- **Priors:**

- $\alpha, \beta \sim \text{Gamma}(0.1, 0.1)$

Beta Mixture Models $H > 1$

Likelihood

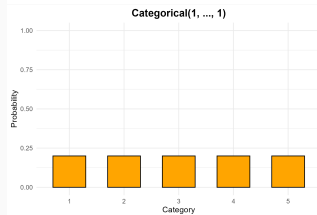
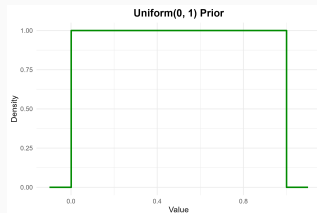
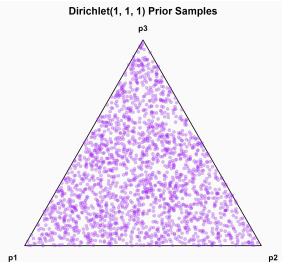
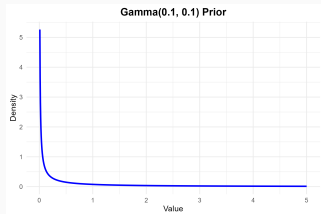
$$H > 1 : p(y_i \mid \alpha, \beta, \pi) = \sum_{h=1}^H \pi_h \text{Beta}(y_i \mid \alpha_h, \beta_h)$$

- **Label Switching Prevention:** Reparameterized in terms of mean (μ_h) and concentration (κ_h).
- Where $\alpha_h = \mu_h \kappa_h$, $\beta_h = (1 - \mu_h) \kappa_h$
- Enforced strict ordering: $\mu_1 < \mu_2 < \dots < \mu_H$.
- **Priors:**
 - $\mu_h \sim \text{Uniform}(0, 1)$
 - $\kappa_h \sim \text{Gamma}(0.1, 0.1)$
 - $\pi \sim \text{Dirichlet}(1, \dots, 1)$

Why Weakly Informative Priors?

- In a mixture model, we do not know a priori the location and spread of the clusters.
- Using a **strongly informative** prior could force the model away from the true modes.
- We opted for a **weakly informative** priors, which allows the data to dominate the posterior.

Why Weakly Informative Priors?



Calculate Log-Likelihood

S	Samples					
	a[1]	...	b[1]	...	pi[1]	...
	0.244		0.267		0.33	
	...		...		...	

N	DataFrame		
	...	...	Load_Factor
			0,3
			0,43
			...

Calculate Log-Likelihood

To calculate the log-likelihood of the orange box we used the following formula:

$$\log \left(\sum_{h=1}^H \pi_h \cdot \text{Beta}(y_i \mid \alpha_h, \beta_h) \right)$$

where $\text{Beta} = \frac{y^{\alpha-1}(1-y)^{\beta-1}}{B(\alpha, \beta)}$ and $B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$

Log Likelihood Matrix			
S		...	
	...		
	...		
	...		
N			

Model	1000 Iter.	2000 Iter.	3000 Iter.
$H = 1$	-1634.36	-1633.79	-1634.11
$H = 2$	-2689.84	-2689.47	-2490.29
$H = 3$	-2800.12	-2988.61	-3098.91
$H = 4$	-2810.93	-3114.57	-3105.56
$H = 5$	-3107.15	-3110.98	-3108.55

- Evaluated models for $H \in \{1, 2, 3, 4, 5\}$.
- $H = 4$ and $H = 5$ show comparable optimal performance. We focus on $H = 4$ for parsimony and ease of interpretability.

Task 2: Posterior Predictive Checks

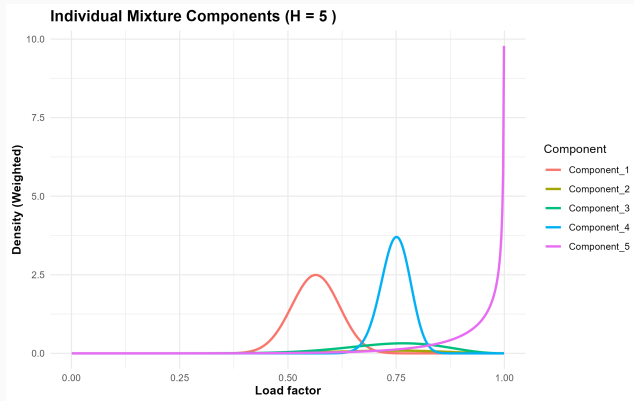
- We generate the predictive density by averaging over the S MCMC samples:

Predictive Density

$$p(\tilde{y} \mid y) \approx \frac{1}{S} \sum_{s=1}^S \sum_{h=1}^H \pi_h^{(s)} \text{Beta}(\tilde{y} \mid \alpha_h^{(s)}, \beta_h^{(s)})$$

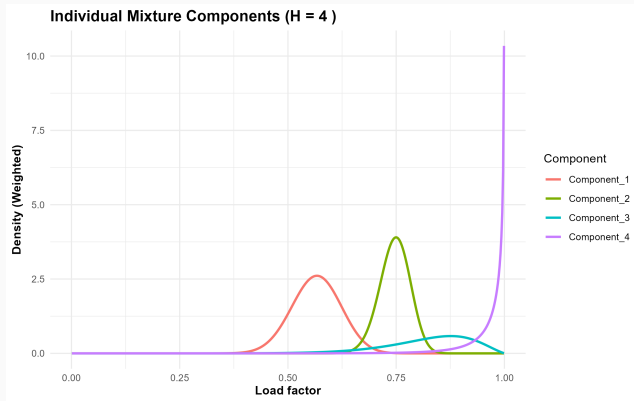
Mixture Sub-Components ($H = 5$)

- **Component 1:** Base load (nightly consumption)
- **Component 2:** Flat
- **Component 3:** High spread component
- **Component 4:** Medium load
- **Component 5:** Peak load (≈ 1)

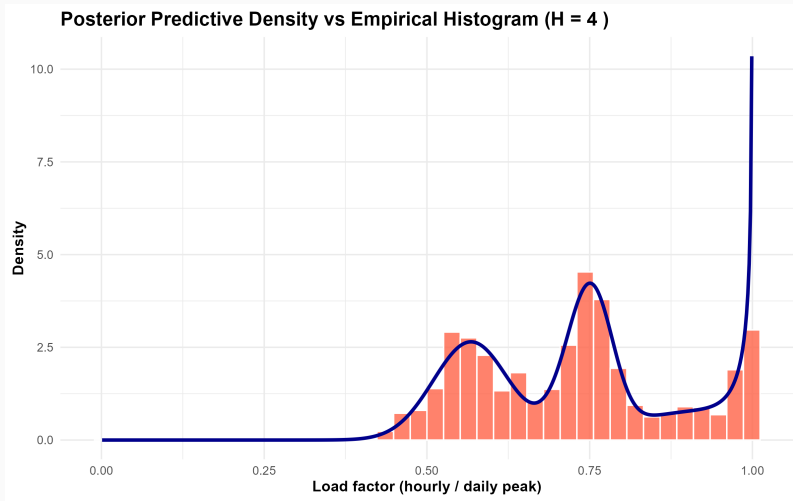


Mixture Sub-Components ($H = 4$)

- **Component 1:** Base load (nightly consumption)
- **Component 2:** Medium load
- **Component 3:** Transition phase (high spread)
- **Component 4:** Peak load (≈ 1)



Mode Recovery



Task 3: Post-Hoc Clustering

MAP Component Assignments

- We calculate the posterior means of the parameters $(\hat{\pi}, \hat{\alpha}, \hat{\beta})$ to classify each observation.

Maximum A Posteriori (MAP) Assignment

$$\hat{P}(Z_i = h \mid y_i) = \frac{\hat{\pi}_h \text{Beta}(y_i \mid \hat{\alpha}_h, \hat{\beta}_h)}{\sum_{k=1}^H \hat{\pi}_k \text{Beta}(y_i \mid \hat{\alpha}_k, \hat{\beta}_k)}$$

- Each observation is assigned to the cluster that maximizes this probability.

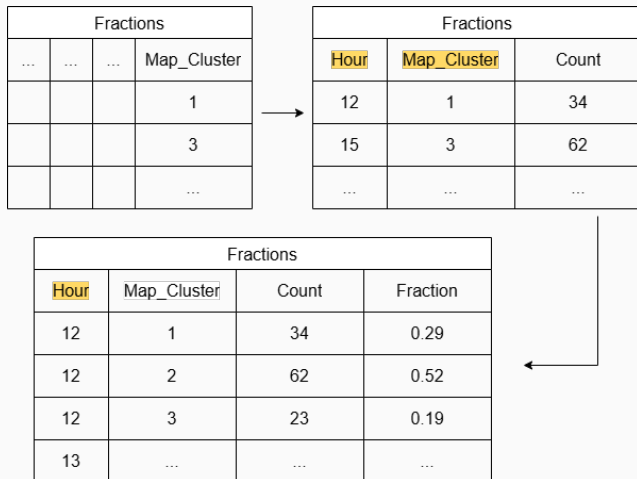
Probability Matrix: $P(Z_i = h \mid y_i)$

N	Probability Matrix			
	0.4	0.1	0.3	0.2
	0.1	0.7	0.05	0.15
	...	...	...	...
H				

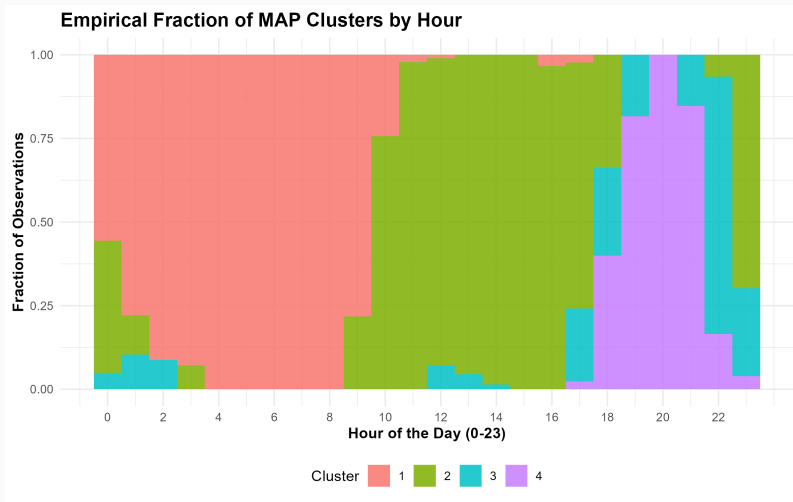
MAP Cluster Assignment

N	DataFrame			
	...	...	...	Map_Cluster
				1
				2
				...

Proposed Pipeline



Temporal Distribution of Clusters



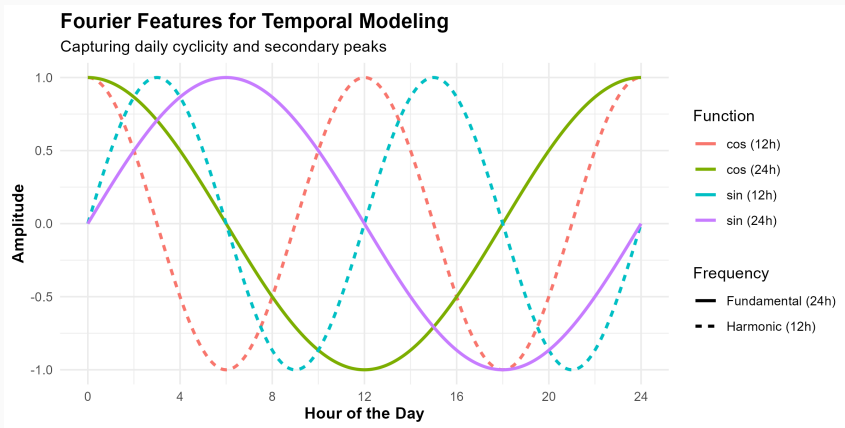
Task 4: Temporal Modeling

The problem with linear time

- **Goal:** Construct a continuous predictive model for cluster assignment.
- **The Issue:** Including the hour linearly (0, 1, 2... 23) forces a discontinuity between 23:59 and 00:00.
- **Solution:** Fourier Transformations.

Fourier Transformations Visualized

These continuous sine/cosine waves act as the new features for our regression.



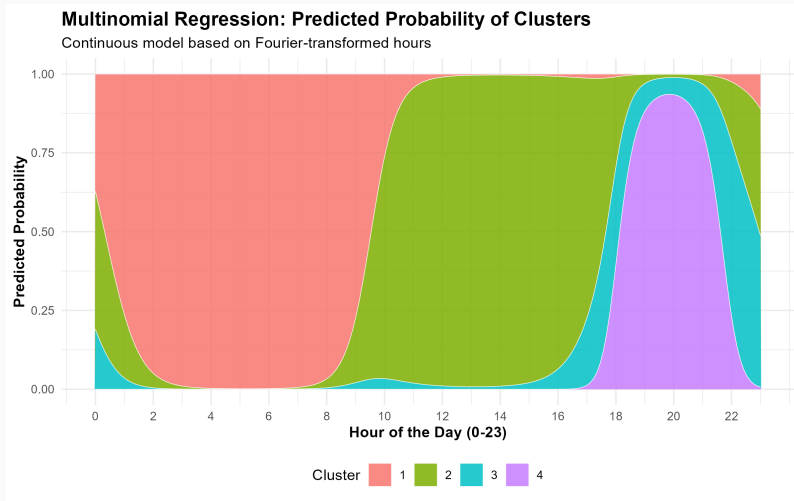
Temporal Multinomial Logistic Regression

- We apply a multinomial logistic regression using the MAP assignments as the response variable.

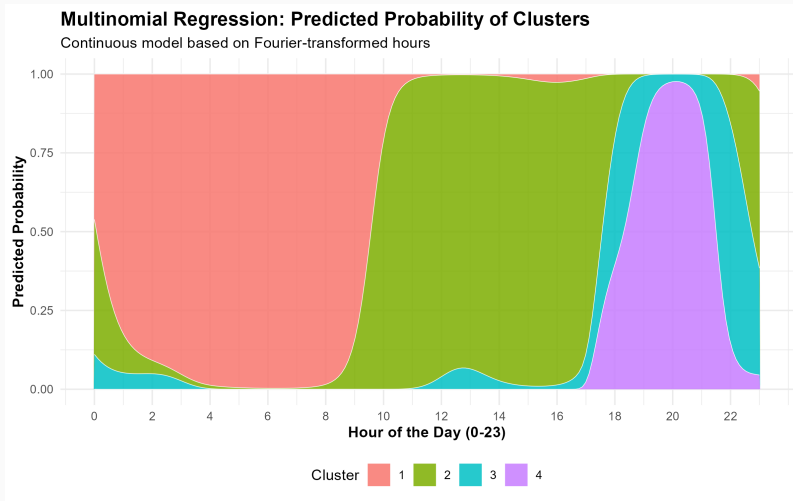
Predictor structure for cluster h

$$\eta_h = \beta_{h,0} + \beta_{h,1} \sin\left(\frac{2\pi \text{hour}}{24}\right) + \beta_{h,2} \cos\left(\frac{2\pi \text{hour}}{24}\right) + \beta_{h,3} \sin\left(\frac{4\pi \text{hour}}{24}\right) + \beta_{h,4} \cos\left(\frac{4\pi \text{hour}}{24}\right)$$

Predicted Probabilities



Why not $K=3$?



Thank you for your attention!